3.6 Using the Quadratic Formula

Name _____ Period ____ Date _____

Algebra II 3.6 (30-minute plan) · Student sheet

I can…

- solve quadratic equations using the Quadratic Formula.
- use the discriminant to find the number and type of solutions.
- use the Quadratic Formula to solve real-life problems.

Vocabulary · fill in the missing word

Quadratic Formula — $x =$ _____, for $ax^2 + bx + c = 0$

Discriminant — _____

$b^2 - 4ac > 0$ — two real _____

$b^2 - 4ac = 0$ — one real _____

$b^2 - 4ac < 0$ — two _____ solutions

Part 1 · Notes and examples (fill in each note, then work the example)

Quadratic Formula: $x =$ _____

Example 1

Solve $x^2 + 3x - 5 = 0$ using the Quadratic Formula.

The discriminant: Discriminant _____ > 0: _____; = 0: _____; < 0: _____.

Example 2

Find the discriminant and describe the solutions.

a. $4x^2 - 12x + 9 = 0$ b. $x^2 + 2x + 7 = 0$ c. $3x^2 + x - 2 = 0$

Part 2 · Partner practice

Partner practice

Solve $x^2 - 6x + 4 = 0$.